

Large Scale Optimization.

Introduction to The Traffic Assignment Problem and Wardrop's Equilibrium Principle

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- **Introduction:**
- **Conceptual definition of Wardrop's First Principle.**
- **Formulation of the Traffic Assignment Problem.**
- **Frank-Wolfe's convex combinations method .**

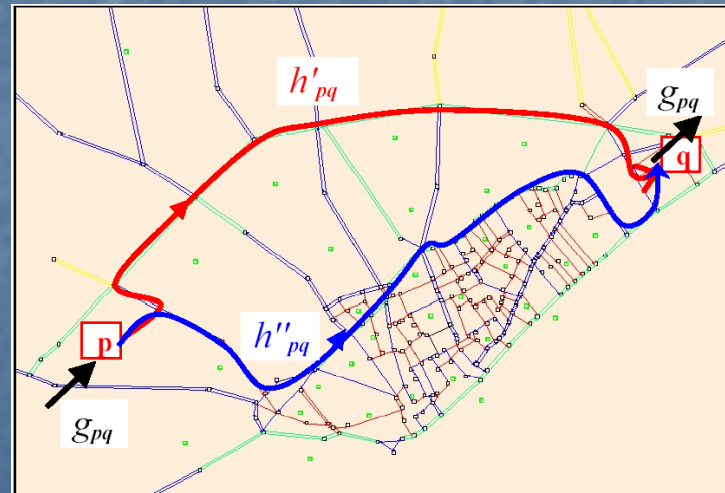


WARDROP's First Principle (User Equilibrium)

In a transportation network, to go from origin p to destination q , routes chosen by travellers are so that total flow $p \rightarrow q$ divides amongst them in such a way that, those resulting with a positive flow have minimum (and thus, equal) cost c_{pq} ;

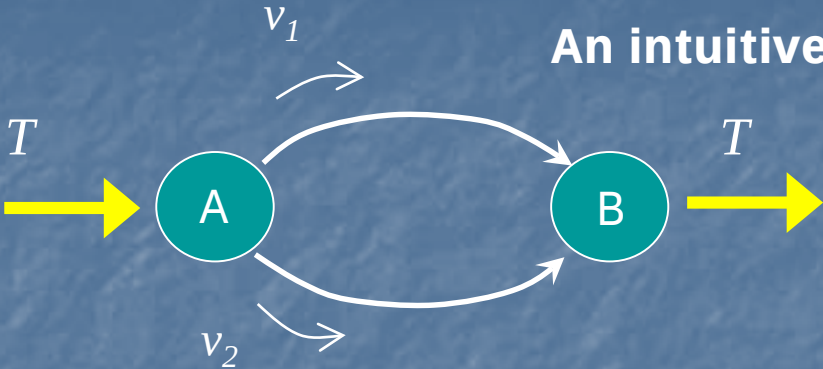
Routes from $p \rightarrow q$ that are non-used are so that

$$\geq c_{pq}$$



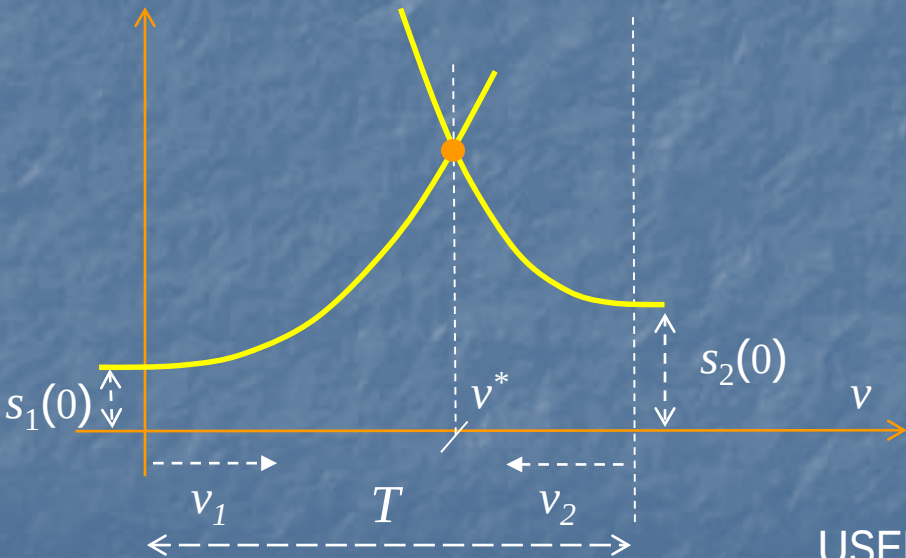
WARDROP'S EQUILIBRIUM PRINCIPLE (1952)

An intuitive example



$$v_1 + v_2 = T, \quad v_1, v_2 \geq 0$$

$$T = 4$$

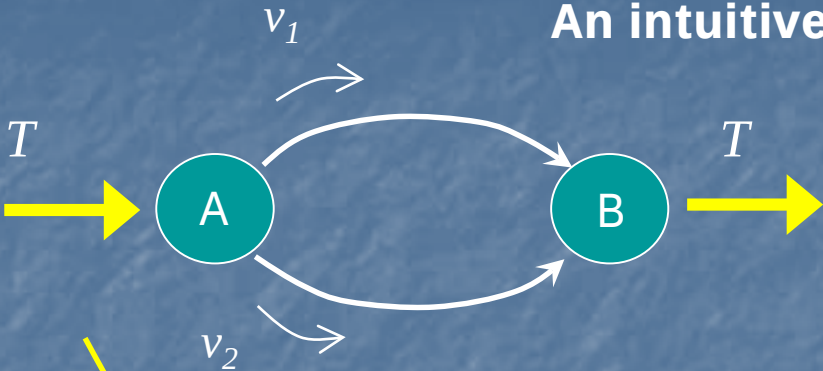


$$s_1(v_1) = 1 + v_1^2, \quad s_2(v_2) = 2 + v_2^2$$
$$s_1(v_1) = s_2(v_2)$$
$$v_1^* = \frac{17}{8}, \quad v_2^* = \frac{15}{8}$$
$$s^* = s_1(v_1^*) = s_2(v_2^*) = \frac{353}{64} = 5,51$$

USER EQUILIBRIUM

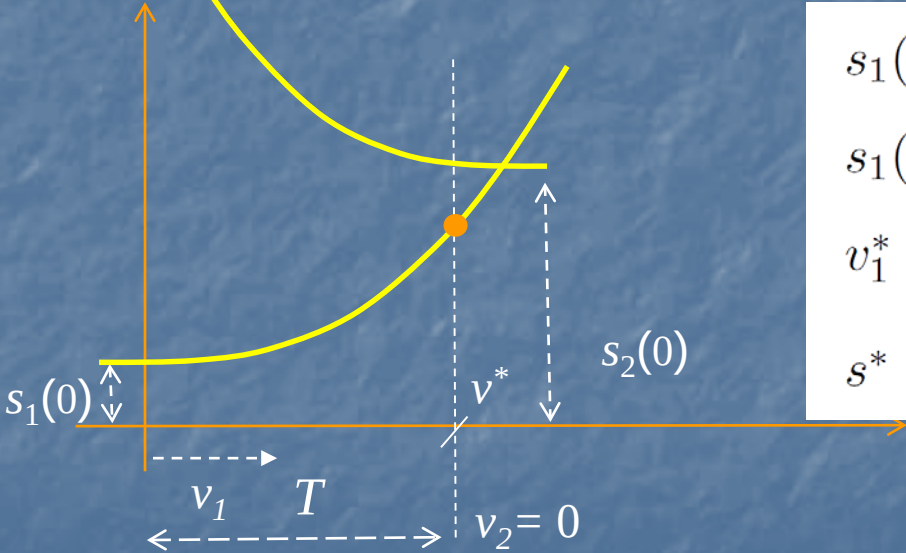
WARDROP'S EQUILIBRIUM PRINCIPLE (1952)

An intuitive example

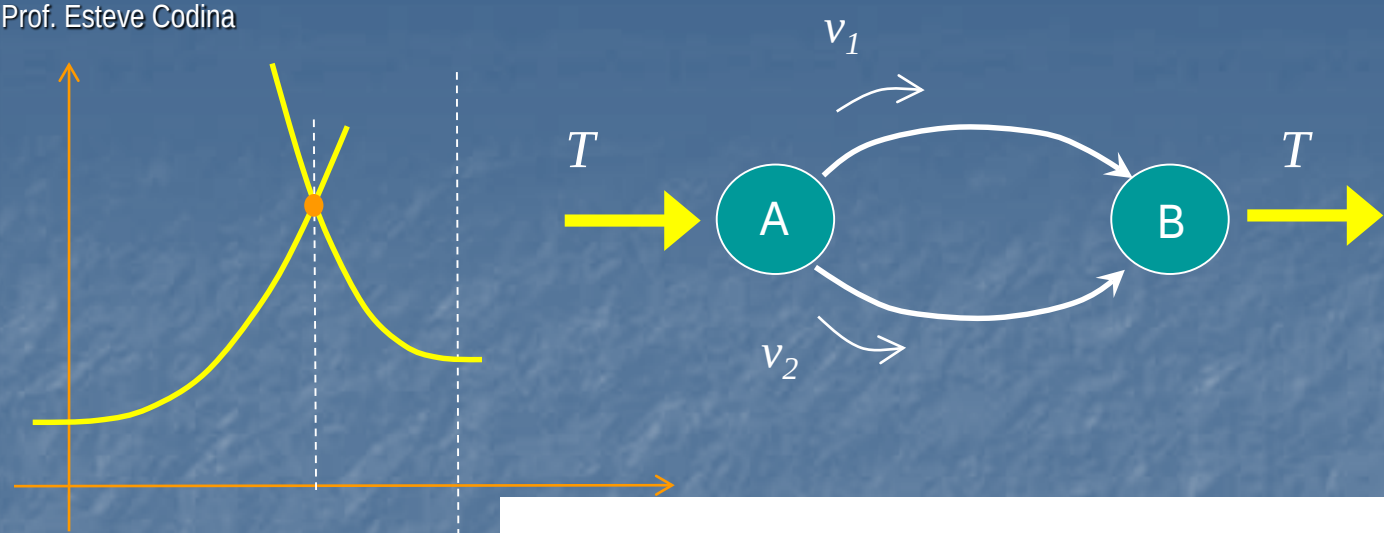


$$v_1 + v_2 = T, \quad v_1, v_2 \geq 0$$

$$T = 1,2$$

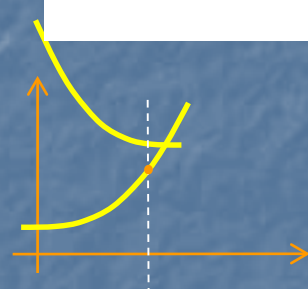


$$s_1(v_1) = 1 + v_1^2, \quad s_2(v_2) = 3 + v_2^2$$
$$s_1(v_1) = s_2(v_2) \text{ sin soluci3n para } v_1 \leq 1,2$$
$$v_1^* = 1,2, \quad v_2^* = 0$$
$$s^* = s_1(v_1^*) = 2,44 < s_2(v_2^*) = 3$$



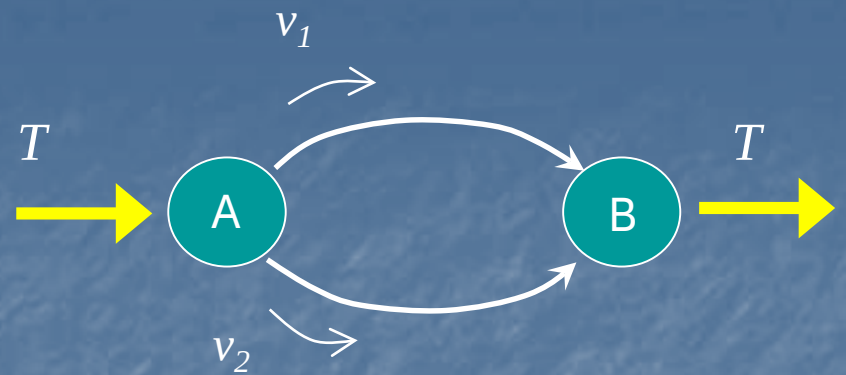
$$\text{Total travel time} = v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*) = T s^* = 4 \frac{353}{64} = 22,06 \text{ h}$$

$$\text{Mean travel time per pax.} = \frac{v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*)}{T} = \frac{T s^*}{T} = s^* = \frac{353}{64} = 5,51 \text{ h-pax}$$



$$\text{Total travel time} = v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*) = T s^* = 2,982 \text{ h}$$

$$\text{Mean travel time per pax.} = \frac{v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*)}{T} = \frac{T s^*}{T} = s^* = 2,44 \text{ h-pax}$$



COOPERATIVE BEHAVIOUR:
SYSTEM-OPTIMUM

Minimizing total travel time Ξ_{Ξ}
Minimize mean travel time (per pax.)

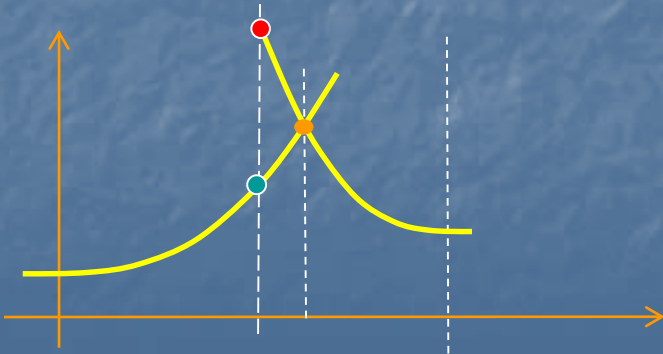
$T = 4$ $v_1^* = 2,042, v_2^* = 1,958$ 

$$\text{Min}_v \quad v_1 s_1(v_1) + v_2 s_2(v_2)$$

$$\text{sujeto a : } \begin{aligned} v_1 + v_2 &= T \\ v_1, v_2 &\geq 0 \end{aligned}$$

Total travel time $= v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*) = 21,979$

Mean travel time per pax. $= \frac{v_1^* s_1(v_1^*) + v_2^* s_2(v_2^*)}{T} = \frac{21,979}{4} = 5,495$



Mean Travel Time is *smaller* than before...
... on behalf of those who chose route 2, a more “expensive” route than route 1

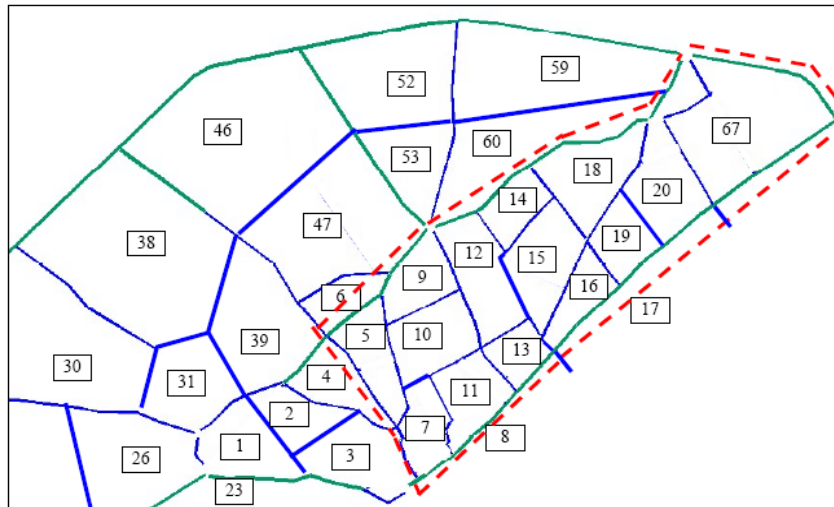
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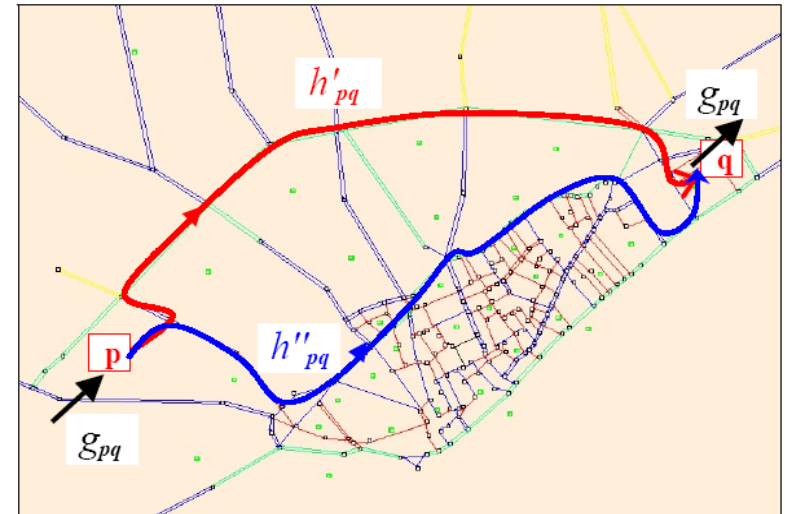
Basic Notation : $G(N,A)$

- N : Set of nodes in the network
- A : Set of links of the network
- W : Origin-Destination pairs
- Γ_w : sub-set of routes connecting o-d pair $w \in W$
- v_a : Flow on link a
- c_a : Travel Cost on link a (costos operación, tiempo, etc.)
- h_p : Flow on route p within Γ_w
- C_p : Travel cost associated to route p
- g_w : Total # of trips for o-d pair $w \in W$

Equilibrium and network flow problems



Zonificación del área



Multicommodity network flows

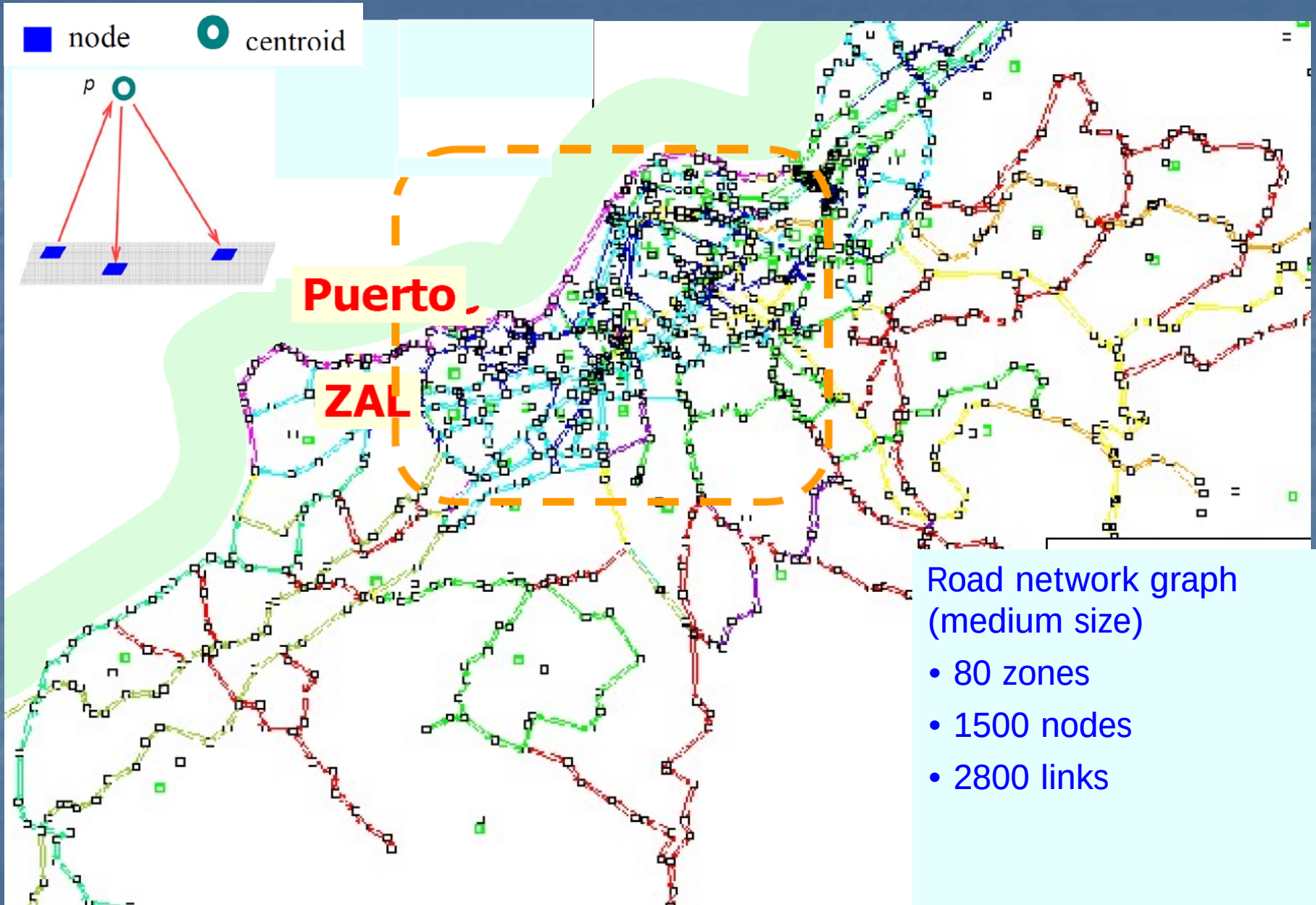
$$\sum_r h_{pq}^r = g_{pq}, \quad h_{pq}^r \geq 0, \quad v_a = \sum_{pq} \sum_{a \in r} h_{pq}^r$$

Wardrop's equilibrium principle

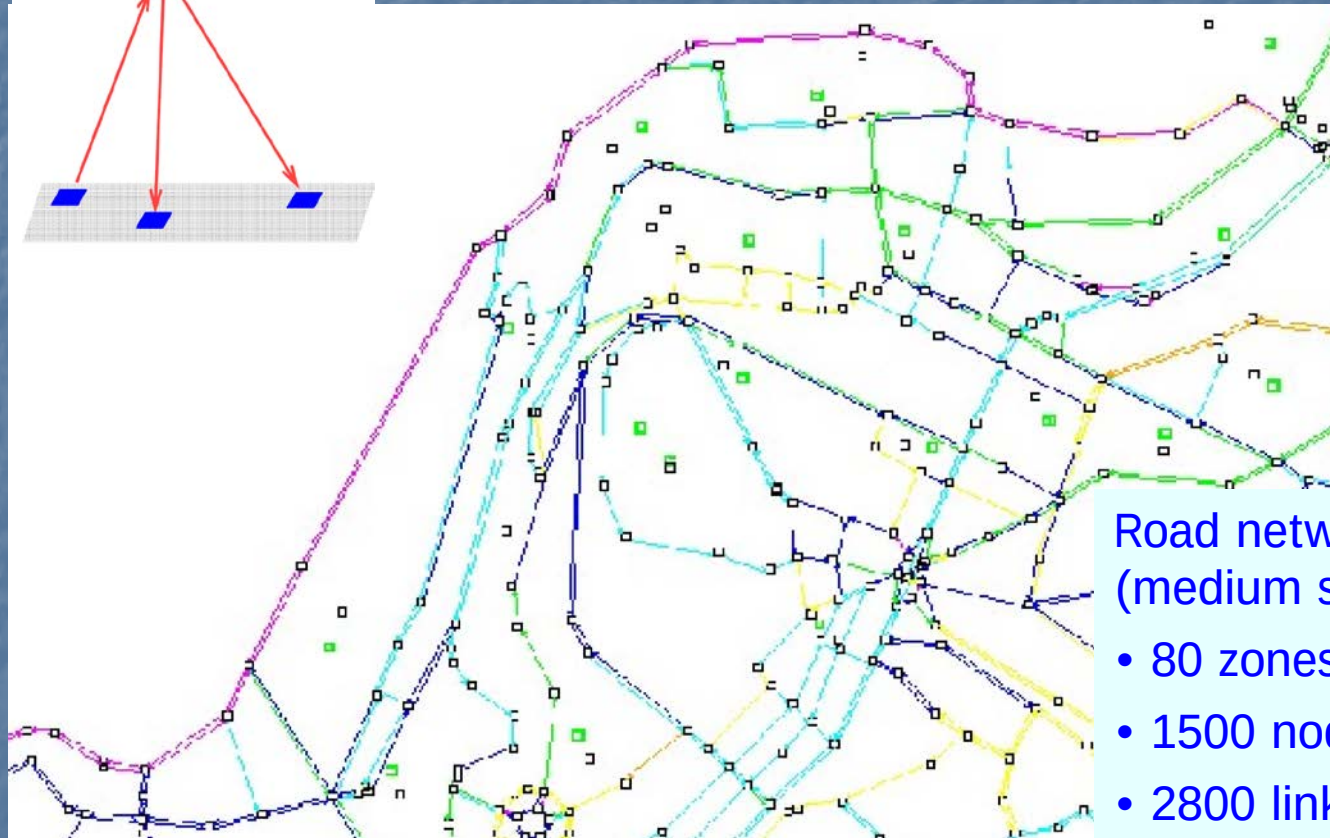
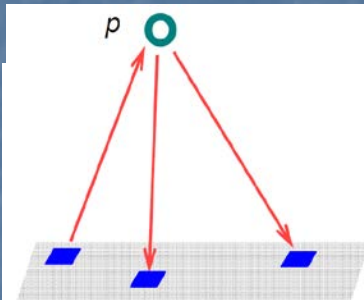
Route's 1 Cost $p \rightarrow q$ = Route's 2 Cost $p \rightarrow q$ = ...

FORMULATION:

- Variational Inequalities in $\mathbb{R}^n$
- Optimization problems on networks



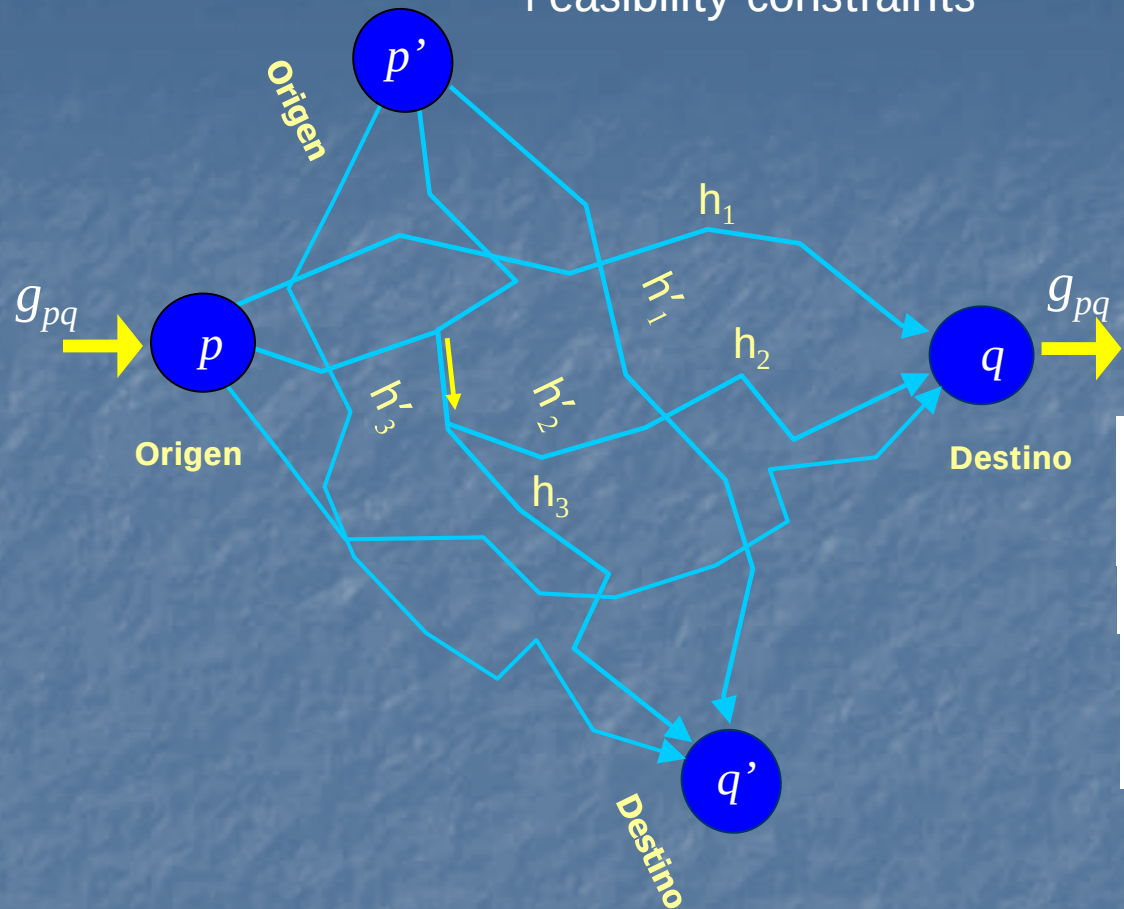
■ node ● centroid



Road network graph
(medium size)

- 80 zones
- 1500 nodes
- 2800 links

Feasibility constraints



$$\omega = (p, q)$$

$$\sum_{r \in \Gamma_\omega} h_r^\omega = g_\omega \quad \omega \in W,$$

$$h_r^\omega \geq 0$$

$$v_a = \sum_{\omega \in W} \sum_{r \in \Gamma_\omega} \delta_{ar} h_r^\omega$$

$$\Omega(g) = \left\{ h \geq 0 ; \sum_{r \in \Gamma_\omega} h_r = g_\omega, \forall \omega \in W \right\}$$

$$\mathcal{V}(g) = \left\{ v \in R^{|A|} ; v = \sum_{\omega \in W} \sum_{r \in \Gamma_\omega} \delta_{ar} h_r^\omega, h \in \Omega(g) \right\}$$

User Equilibrium (Wardrop).

Mathematical Formulation:

$$\Omega(g) = \left\{ h \geq 0 ; \sum_{r \in \Gamma_w} h_r = g_w, \forall w \in W \right\}$$

Feasible flows on paths

$$\forall w \in W, C_r^\omega \begin{cases} = u_\omega^* \leftarrow h_r^\omega > 0 & \forall r \in \Gamma_\omega, \\ > u_\omega^* \rightarrow h_r^\omega = 0 & \forall r \in \Gamma_\omega, \end{cases}$$

$$u_\omega^* = \min_{r \in \Gamma_\omega} \{ C_r^\omega \}$$

Previous conditions are equivalent to:

$$\left. \begin{aligned} h_r^\omega (C_r^\omega(h) - u_\omega^*) &= 0 \\ C_r^\omega(h) - u_\omega^* &\geq 0 \end{aligned} \right\} \forall r \in \Gamma_\omega, \omega \in W$$

Complementarity

$$\sum_{\omega \in W} \sum_{r \in \Gamma_\omega} C_r^\omega(h^*) (h_r^\omega - h_r^{\omega,*}) \geq 0$$

$$\forall h \in \Omega(g)$$

Variational Inequality

User Equilibrium (Wardrop).

Smith's mathematical formulation (1979)

From the previous complementarity, there follows the V.I. :

Find $h^* \in \Omega(g)$ so that

$$\sum_{\omega \in W} \sum_{r \in \Gamma_{\omega}} C_r^{\omega}(h^*)(h_r^{\omega} - h_r^{\omega,*}) \geq 0$$

$$\forall h \in \Omega(g)$$



Find $v^* \in \mathcal{V}(g)$ so that

$$\sum_{a \in A} s_a(v)(v_a - v_a^*) \geq 0$$

$$\forall v \in \mathcal{V}(g)$$

The diagonal traffic assignment problem with inelastic demand

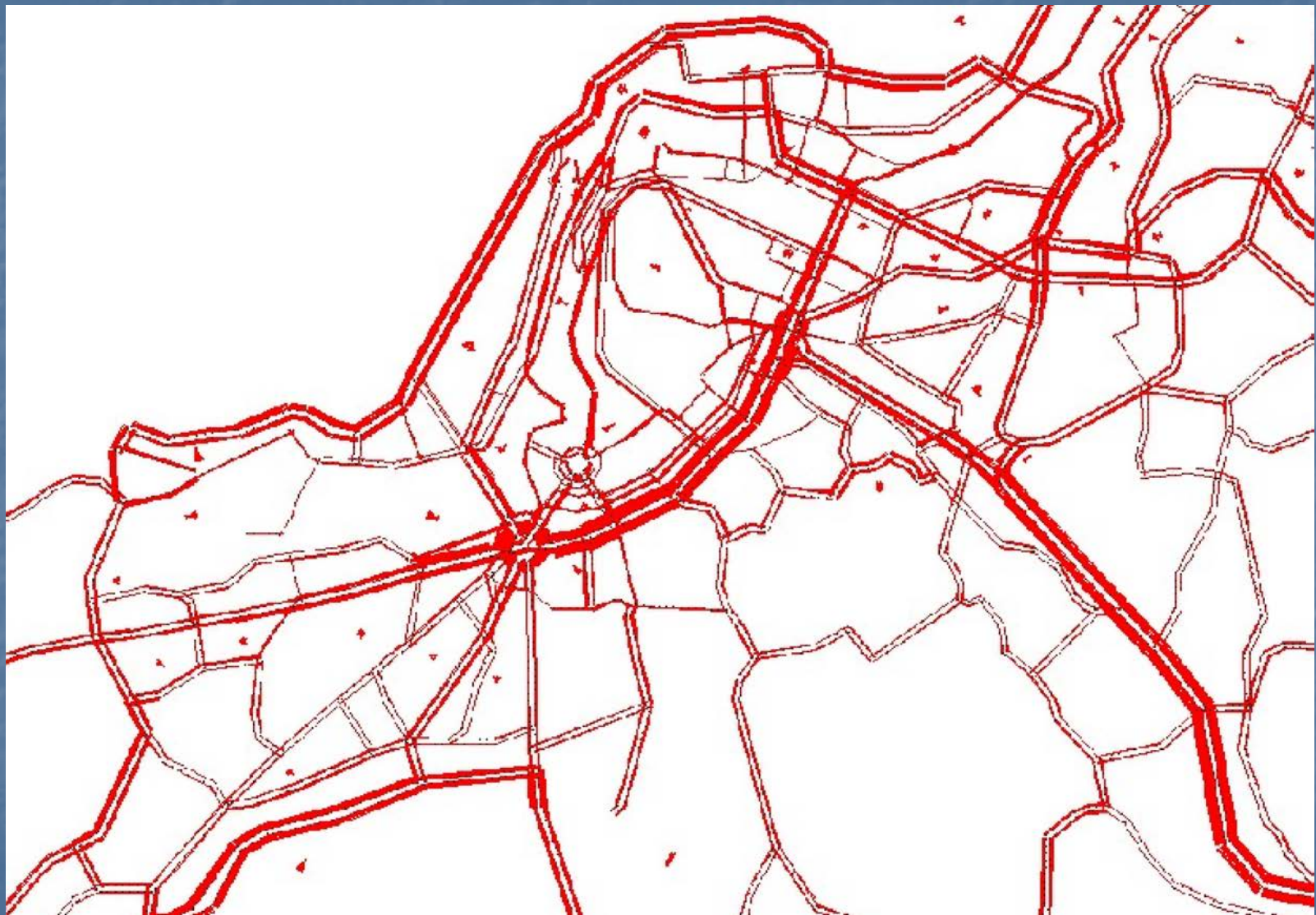
$$\begin{aligned}
 & \text{Min}_v \quad \sum_{a \in A} \int_0^{v_a} s_a(x) dx \\
 & \text{sujeto a :} \quad \sum_{r \in \Gamma_\omega} h_r^\omega = g_\omega \quad \omega \in W \\
 & \quad \quad \quad h_r^\omega \geq 0 \\
 & \quad \quad \quad (v_a = \sum_{\omega \in W} \sum_{r \in \Gamma_\omega} \delta_{ar} h_r^\omega) \quad a \in A
 \end{aligned}$$

$$\omega = (p, q)$$

$$\text{Min}_{v \in \mathcal{V}(g)} \quad \sum_{a \in A} \int_0^{v_a} s_a(x) dx$$

$$\mathcal{V}(g) = \left\{ v \in R^{|A|} ; v = \sum_{\omega \in W} \sum_{r \in \Gamma_\omega} \delta_{ar} h_r^\omega, \quad h \in \Omega(g) \right\}$$

Results from an assignment: traffic volumes on links:



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Frank-Wolfe's method (1956)

1. **Init:** Let v^0 the initial solution inicial obtained by an all-or-nothing assignment of the o-d matrix demand g on minimum-cost paths (empty network) $s^0 = s(0)$.
2. **Update link costs on links:** $k = k+1$; $s^k = s(v^{k-1})$.
3. **Descent direction :** volumes y^k are obtained by the all-or-nothing assignment with costs s^k .
4. **Optimal step length calculation** λ^k ,
 - $(0 \leq \lambda^k \leq 1)$, the objective function is minimkized along direction $y^k - v^{k-1}$.
5. **Update :** $v^k = v^{k-1} + \lambda^k (y^k - v^{k-1})$.
6. **Stopping Criterion.** Si $|s^k \top (v^{k-1} - y^k)| > \varepsilon$ return to step 1.
 Otherwise **STOP.** $v^* = v^k, s^* = s(v^k)$.

FW-Algorithm for Beckmann's problem

- 0) Initialize $\nu = 0$. Let $v^{(0)}$ be a vector of feasible flows ;
 $BLB^{(0)} = -\infty$
- 1) Linearization of objective function $C(v)$ at $v^{(\nu)}$ and "subproblem" stage

$$\left. \begin{array}{l} \text{Min}_{x^k} \sum_{k \in \mathcal{K}} \sum_{a \in \mathcal{A}} \gamma_a^k x_a^k \\ \text{s.t. } Bx^k = d^k \\ x^k \geq 0, \quad k \in \mathcal{K} \end{array} \right\} \rightarrow \hat{v}_a^{(\nu)} = \sum_{k \in \mathcal{K}} \hat{v}_a^{k,(\nu)} \quad (SP)$$

$$\gamma_a^k = c_a(v_a) = \frac{\partial C}{\partial v_a^k} \left(\sum_{k \in \mathcal{K}} v_a^{k,(\nu)} \right)$$

use the descent direction $d_a^{(\nu)} = \hat{v}_a^{(\nu)} - v_a^{(\nu)}$

Update the "best lower bound" BLB :

$$BLB^{(\nu+1)} = \max\{BLB^{(\nu)}, C(v^{(\nu)}) + \nabla_v C(v^{(\nu)})^T (\hat{v}_a^{(\nu)} - v^{(\nu)})\}$$

2) (Line search)

$$\text{If } \phi(\lambda) = \sum_{a \in \mathcal{A}} \int_0^{v_a^{(\nu)} + \lambda d_a^{(\nu)}} c_a(\omega) d\omega = \sum_{a \in \mathcal{A}} C_a(v_a^{(\nu)} + \lambda d_a^{(\nu)})$$

$$\text{Find } \hat{\lambda}^{(\nu)} = \operatorname{argmin}\{\phi(\lambda) \mid 0 \leq \lambda \leq 1\}$$

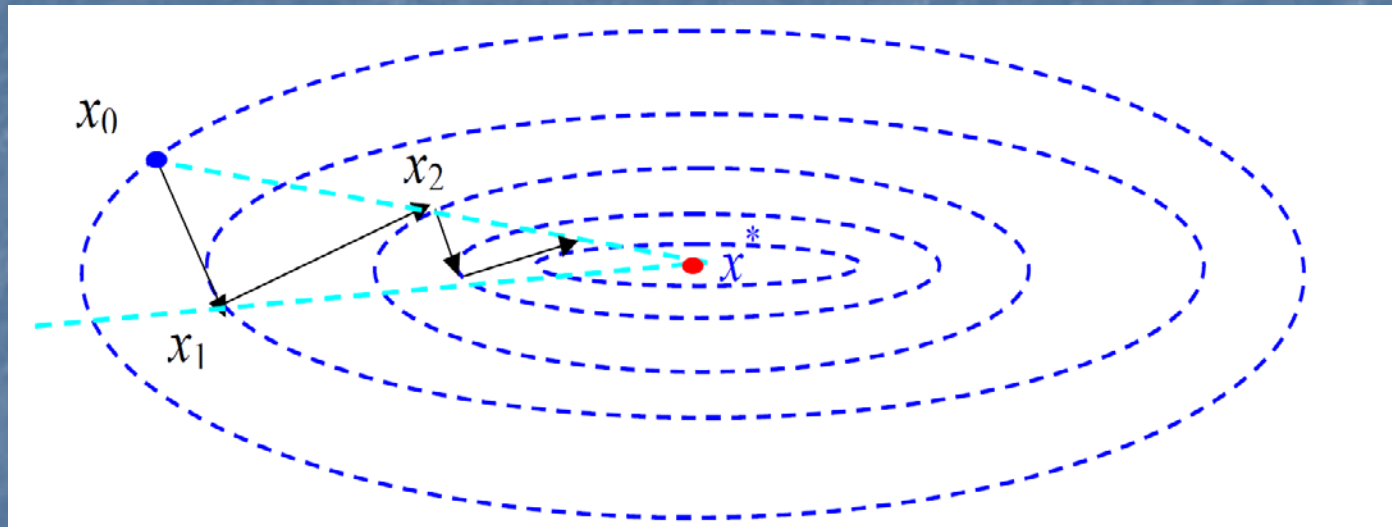
$$v_a^{(\nu+1)} = v_a^{(\nu)} + \hat{\lambda}^{(\nu)} d_a^{(\nu)}$$

$$\text{If } \frac{C(v_a^{(\nu+1)}) - BLB^{(\nu+1)}}{BLB^{(\nu+1)}} \leq \epsilon \quad \text{then } STOP.$$

$$\text{otherwise } \nu \leftarrow \nu + 1; \text{ goto step 1)}$$

F-W's behaviour

Although FW converges theoretically to a solution, the practical implementations cannot reduce the gap to a very small value $\rightarrow$ TAP cannot be solved with very high accuracy.



- 0) Initialize $\nu = 0$. Let $x^{(0)} \in X$.
 $BLB = -\infty$, $W_x^{(0)} = \{x^{(0)}\}$, $W_s^{(0)} = \emptyset$
- 1) ("Subproblem") Linearization of the objective function at $x^{(\nu)}$.
 A new vertex is obtained

$$\left. \begin{array}{l} \text{Min}_x \quad \nabla_x f(x^{(\nu)})^T x \\ \text{s.t.} \quad x \in X \end{array} \right\} \rightarrow \hat{x}^{(\nu)} \quad (SP)$$

2) (Updating the working sets)

If $|W_s^{(\nu)}| < \rho$

a new vertex is stored: $W_s^{(\nu+1)} = W_s^{(\nu)} \cup \{\hat{x}^{(\nu)}\}, \quad W_x^{(\nu+1)} = W_x^{(\nu)}$

If $|W_s^{(\nu)}| = \rho$

Drop the element in $W_s^{(\nu)}$ with smaller baricentric coordinate $\tilde{x}$,
 add the new vertex $\hat{x}^{(\nu)}$: $W_s^{(\nu+1)} = W_s^{(\nu)} - \{\tilde{x}\} \cup \{\hat{x}^{(\nu)}\}$ and
 update $W_x^{(\nu+1)} = \{x^{(\nu)}\}$

New set of extreme points for the "master problem"

$$W^{(\nu+1)} = W_s^{(\nu+1)} \cup W_x^{(\nu+1)}$$

3) (Stopping Criterion) Updating of the best lower bound BLB :

$$BLB^{(\nu+1)} = \max\{BLB^{(\nu)}, f(x^{(\nu)}) + \nabla_x f(x^{(\nu)})^T (\hat{x}^{(\nu)} - x^{(\nu)})\}$$

$$\text{If } \frac{f(x^{(\nu)}) - BLB^{(\nu+1)}}{BLB^{(\nu+1)}} \leq \epsilon \text{ then } STOP.$$

otherwise go to step 4)

4) ("Master Problem")

$$\left. \begin{array}{l} \text{Min}_x \quad f(x) \\ \text{s.t.} \quad x \in \text{CONV}\{W^{(\nu+1)}\} \end{array} \right\} \rightarrow x^{(\nu+1)} \quad (MP)$$

Columns from $W_s^{(\nu+1)}$ and $W_x^{(\nu+1)}$
with null baricentric coordinate are
eliminated

$\nu \leftarrow \nu + 1$; goto step 1)

	Zonas	Nodos	Arcos	nOD
SIO	24	24	76	538
BCN	110	930	2522	7922
WIN	154	1017	2976	4345
CMB	90	2253	5171	8004
MAD	490	2776	6871	185000

E.Codina, L.Montero, J.Barceló
"A Computational Comparison of
two Simplicial Decomposition
Approaches for the Separable Traffic
Assignment Problem: RSDTA and
RSDVI."
DR 1996/05 EIO

RSDTA Computational Results (STOPPING GAP 1%) NO INTERACTION TESTS							
NET WORK	Maxver	Low Congestion			High Congestion		
		CPUTotal	CPUMp(%)	Niter	CPUTotal	CPUMp(%)	Niter
SIO	4	0.969	50.65	6	3.212	63.57	21
SIO	8	0.911	54.43	(4) 6	2.747	67.64	15
SIO	12	0.913	54.77	(4) 6	2.731	67.68	(8) 15
SIO	15	0.906	54.20	(4) 6	2.735	67.65	(8) 15
SIO	20	0.908	54.36	(4) 6	2.729	67.69	(8) 15
SIO	30	0.908	54.36	(4) 6	2.729	67.69	(8) 15
SIO	Unlim.	0.909	54.31	(4) 6	2.732	67.63	(8) 15
BCN	4	366.6	15.99	34	1182.9	17.98	121
BCN	8	277.2	22.89	25	1201.9	26.83	109
BCN	12	274.6	25.12	(10) 24	1087.6	34.69	87
BCN	15	275.0	24.98	(10) 24	1105.4	39.27	82
BCN	20	274.8	25.10	(10) 24	1252.7	49.23	78
BCN	30	275.6	25.16	(10) 24	1156.7	45.29	(23) 78
BCN	Unlim.	275.5	25.15	(10) 24	1156.6	45.29	(23) 78
WIN	4	197.6	19.92	22	1199.9	24.70	156
WIN	8	207.8	28.08	21	1270.5	36.02	140
WIN	12	178.1	26.79	(9) 18	1403.0	45.86	131
WIN	15	178.5	26.85	(9) 18	1556.3	52.51	127
WIN	20	178.2	26.91	(9) 18	1688.1	60.83	113
WIN	30	178.2	26.91	(9) 18	1688.1	60.83	113
WIN	Unlim.	178.2	26.93	(9) 18	1805.5	63.09	(28) 114
CMB	4	1033.8	28.35	82	3644.1	29.24	284
CMB	8	1188.7	40.02	80	4570.8	40.64	295
CMB	12	1421.7	49.62	80	4999.9	51.43	269
CMB	15	1609.4	55.63	79	5809.8	58.13	273
CMB	20	1849.5	62.80	76	7279.7	66.75	271
CMB	30						
CMB	Unlim.	2063.9	67.47	(30) 74	15377.0	86.07	(49) 239
MAD	4	1492.5	3.22	12	9843.9	4.52	96
MAD	8	1609.4	4.28	(9) 13	10492.8	7.49	98
MAD	12	1608.9	4.32	(9) 13	9924.7	10.68	89
MAD	15	1607.9	4.30	(9) 13	10681.6	13.48	92
MAD	20	1611.4	4.31	(9) 13	10021.5	17.01	82
MAD	30	1609.5	4.32	(9) 13	9794.8	17.86	(25) 81
MAD	Unlim.	1609.5	4.32	(9) 13	9794.8	17.86	(25) 81

